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Mattia Garatti

Curriculum Vitae

I am a doctoral researcher in Mathematics interested in Mathematical Analysis. Currently, I am carrying out my research on regularized optimal transport problem(s). I am also a private teacher, mainly for high school and university students.

Current position

Doctoral researcher in Mathematics, *Université Paris-Saclay*, Orsay, France. **2025 – ongoing**
Thesis: "On regularized Optimal Transport problems". Supervisors: Luca Nenna - Simona Rota-Nodari.

Education

Master degree in Mathematics (LM-40), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2023 – 2025**
Average: 30 cum laude. Main courses' topics: Real and Functional Analysis, Measure Theory, Differential Geometry, Mathematical Physics. Thesis: "Variational representations of optimal transport and Schrödinger problem". Supervisor: Luca Tamanini - Cosupervisor: Nicolò De Ponti.

Bachelor degree in Mathematics (L-35), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2020 – 2023**
Average: 31.33/30. Thesis: "Il Teorema di esistenza di Peano per le equazioni differenziali ordinarie", provision of a theory on ODEs in the Peano's assumptions. Supervisor: Marco Marzocchi

High School Diploma, *Liceo Scientifico 'Camillo Golgi'*, Breno (BS), 100/100. **2015 – 2020**

Publications

A PDE approach to Benamou–Brenier formula for the Schrödinger problem, *M. Garatti, L. Nenna, S. Rota Nodari, L. Tamanini*, pre-print. **2026**

Talks

Entropic regularization(s) of Optimal Transport problem(s), *Università Cattolica del Sacro Cuore*, Brescia. **March, 2026**
Talk about Schrödinger problem with a focus on the Benamou–Brenier formula

Variational representations of optimal transport and Schrödinger problem, *Math PhD Days at UniTO*, Turin. **February, 2025**
Talk on my master thesis work

Research visits

Université Cote d'Azur, *Nice*. **March 16 - March 27, 2026**
Visited Simona Rota-Nodari

Università Cattolica del Sacro Cuore, *Brescia*. **March 05 - March 06, 2026**
Visited Luca Tamanini

Università Cattolica del Sacro Cuore, *Brescia*. **January 30 - February 06, 2026**
Visited Luca Tamanini

INRIA, *Paris (France)*. **July 28 - August 01, 2025**
Visited Luca Nenna

Teaching

Teaching Assistant, *Université Paris-Saclay*, Orsay, France. **2026 – ongoing**
Conducted problem sessions (TD) for undergraduate students (64 hours)

Tutor, *Università Cattolica del Sacro Cuore*, Brescia, Italy. **June 2025**
Conducted problem sessions for the "MathExcellence 2025" advanced mathematics program (5 hours)

Teaching Intern, *Liceo Scientifico 'G. Golgi'*, Breno, Italy. **September 2022**
Delivered lectures to high school students, administered/graded tests and supported lab activities

Tutor, *Cooperativa 'Assolo'*, Berzo Inferiore, Italy. **August – September 2018**
Provided targeted Mathematics and Physics lessons for students with SLD

Teaching Intern, *Fondazione Scuola Materna 'San Giuseppe'*, Gratacasolo, Italy. **June 2018**
Supervised preschool children and organized early science experiments

Work experience

Math writer, Zanichelli editore S.p.A., Bologna (BO). **2023 – 2025**
Teamwork with the editorial staff, autonomous creation of exercises of various difficulty on different topics to be included in future publications

Summer schools and workshop attended

Journées SMAI-MODE, Nice. **March 2026**
Conférence biennale du groupe MODE de la Société de Mathématiques Appliquées et Industrielles

Modelling Viscoelastic Fluids, *Università Cattolica del Sacro Cuore*, Brescia. **February 2026**
Lecture series about mathematical models for viscoelastic fluids

Calculus of Variations and applications, *CIRM*, Marseille, France. **January 2026**
Workshop on the most recent advances in the field

Geometry, duality and convexity in new OT problems, *Université Paris-Saclay*, Orsay, France. **October 2025**
Workshop on new Optimal Transport problems arising in various applications

PGMO Days 2025, *EDF Lab Paris–Saclay*, Palaiseau, France. **October 2025**
Workshop on Optimal Trasport and its interactions with Data Sciences

Advances in Calculus of Variations and Nonlinear Analysis, *Università Cattolica del Sacro Cuore*, Brescia. **September 2025**
Workshop in honor of Marco Degiovanni's 70th birthday

Il Problema di Plateau, *Università Cattolica del Sacro Cuore*, Brescia. **April 2025**
Lecture series about different approach to Plateau problem(s)

Introduction to Mechanics of Swelling, *Università Cattolica del Sacro Cuore*, Brescia. **February 2025**
Lecture series about mathematical models for swelling

Introduzione ai modelli matematici per i cristalli liquidi, *Università Cattolica del Sacro Cuore*, Brescia. **February 2024**
Lecture series about mathematical models for liquid crystals

Interuniversity Mathematics Summer School, *University of Perugia*. **July – August 2023**
Complex Analysis and PDE courses with related final exams successfully passed

Mathematics in Ljubljana, *University of Ljubljana, FMF and IMFM*. **July 2023**
Summer School with courses on Formalization of Mathematics and Introduction to Complex Dinamics

Awards

"Smart" study award, *Università Cattolica del Sacro Cuore*, Brescia (BS). **2025**
Award for 100 best students of the university

Merit scholarship, *Tenaris*, Dalmine (BG). **2017 – 2025**
Scholarships awarded on merit every year during high school and university

Merit scholarship, *Consorzio del Bacino Imbrifero Montano dell'Oglio*, Bergamo. **2021 – 2025**
Scholarships awarded on merit every year during university

Exemption from first year university fees, *Università Cattolica del Sacro Cuore*, Brescia. **2020**
Exemption by merit

Merit scholarship, *Comune di Costa Volpino* (BG). **2020**
Scholarships awarded on merit for high school diploma

Organization

Academic roles

GdT OT-EDP-ML Seminar, Université Paris-Saclay, Orsay, France. **2026 – ongoing**
Co-organizer of the monthly research seminar on Optimal Transport, PDEs, and Machine Learning

Joint Teachers–Students Commission, Università Cattolica, Brescia. **2024–2025**
Student representative for the evaluation of quality assurance mechanisms as part of the university governance body

Outreach & Community Involvement

Fête de la Science, Université Paris-Saclay, Orsay. **2026**
Staff member for a stand on polyhedra for the primary school

Galileo Prize. **2024**
Students' Jury Member of the prize

Celebration of Mind, Notte dei Ricercatori, Venite a vedere la Terra che gira, Università Cattolica del Sacro Cuore. **2023 – ongoing**
Staff member for the science outreach and mathematical popularization events

Disfida Matematica, Brescia. **2023 – ongoing**
Staff member for the national high school team math competition

Polling Station, *Comune di Costa Volpino (BG)*. **2021–2025**
Scrutineer and polling station president

Parish Volunteer, *Parrocchia di "San Zenone"*, Gratacasolo. **2015 – ongoing**
Summer camp organizer and choir member (with budget managing functions)

Computer skills

Programming languages: Python, C, C++

Mark-up languages: LaTeX, HTML

Languages

Italian: Mother tongue

English: Fluent

French: Fluent

Miscellanea

Worked as a background actor in "L'âge de la solitude"

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October 6th, 2026



Mattia Garatti